# BIOPHYSICAL MODELING, MACHINE LEARNING PREDICTION, AND MOLECULAR DOCKING ANALYSIS FOR DENGUE DISEASE DYNAMICS

**BSC project Proposal**

Submitted to
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# 1 Introduction

## 1.1 Background

Dengue is a viral disease spread by mosquitoes, mainly *Aedes aegypti* [1]. It has become a big health issue in Nepal. In the past, this disease was only seen in hot tropical countries, but now it is common in the Kathmandu Valley too. The first case in Nepal was found in 2004 in Chitwan [8]. Since then, we have seen large outbreaks, especially in 2019 and 2022 [3].

An illustration of the study context is shown in Figure 1, representing the central landmark of our research environment.

Figure 1: Tri-Chandra Multiple Campus (Ghantaghar), the center of this research.

Many people think Dengue spreads randomly. However, from a physics view, it is not random. It works like a machine driven by heat [9]. The mosquito needs a specific temperature to survive and spread the virus. If the temperature is too low (below 18°C), the mosquito cannot function [4]. Because of climate change and urbanization, Kathmandu is getting warmer, which allows these mosquitoes to breed here [8].

## 1.2 Problem Statement

Right now, the government and hospitals mostly count cases after people get sick. This is called statistical surveillance [1]. The problem is that it is reactive. It tells us what happened yesterday, not what will happen tomorrow.

Also, most mathematical models used to study Dengue are too simple. They assume that if a mosquito bites you, you get sick immediately [2]. In reality, there is a delay. The virus takes time to grow inside the mosquito and inside the human. This is called the ”Incubation Period” [10]. Standard models ignore this time lag. There is a need for a system that uses Physics (to understand the mechanism) and Computer Code (to process data) to predict these outbreaks before they become disasters.

### 1.3 Objectives

The main goal of this project is to build a computational model to predict Dengue outbreaks in Kathmandu.

1. To build a **Biophysical SEIR-Vector Model** that includes the biological time lags (incubation).
2. To simulate how **Control Strategies** (like spraying or drugs) can stop the spread.
3. To use **Machine Learning** (Random Forest) on NASA weather data to predict future cases [5].
4. To use **Molecular Docking** to check if certain chemical compounds can bind to the Dengue virus protein [6].

# 2 Literature Review

## 2.1 Early Math Models (1927)

The idea of using math to track diseases started a long time ago. In 1927, two scientists named Kermack and McKendrick came up with the **SIR Model** (Susceptible-Infected-Recovered) [2]. They showed that an epidemic ends when the number of healthy people drops below a certain level. This was a great discovery, but their model was too simple. It assumed people infect each other directly, like the flu. It did not account for mosquitoes or the time it takes for the virus to grow inside the body. My project fixes this by adding a "Vector" (Mosquito) part to the equations [10].

## 2.2 Machine Learning Studies (2019)

Recently, researchers have started using computers to predict diseases. A study by Benedum and colleagues (2019) used a method called **Random Forest** to predict Dengue outbreaks in Peru [5]. They found that this AI method was much better than traditional statistics because it could handle complex weather patterns. However, they only looked at the data; they didn't look at the physics of how the disease actually spreads. My project combines both: the AI prediction and the physical model.

## 2.3 Temperature and Mosquitoes (2001)

A study by Hopp and Foley (2001) proved that mosquitoes are very sensitive to temperature [4]. They showed that the *Aedes* mosquito works best between 18°C and 32°C. If it gets colder than that, the mosquito stops working. This study gives me the scientific proof to use Temperature as the main input for my model.

## 2.4 The Situation in Nepal (2020)

In Nepal, a study by Adhikari and Supakankunti (2020) looked at the big Dengue outbreak of 2019 [3]. They showed that the disease has moved up from the lowlands (Terai) to the hills (Kathmandu) because the city is getting warmer and more crowded. Their study focused on the money lost due to the disease. My project is different because it focuses on predicting when the next wave will hit.

## 2.5 Computer-Aided Drug Finding (2010)

For the drug discovery part, I am using a tool called **AutoDock Vina**. This software was created by Trott and Olson (2010) [6]. They invented a faster way to calculate how well a drug molecule sticks to a protein. This helps us test thousands of potential medicines on a computer without needing a lab [7].

### 2.6 Climate Change in Nepal (2015)

A crucial study by Dhimal et al. (2015) looked specifically at the link between climate change and diseases in the Himalayas [8]. They analyzed data from the Nepal Health Research Council and found a direct linear relationship between the warming climate and the spread of vector-borne diseases into the mountain regions. This supports my hypothesis that the 2022 outbreak in Kathmandu (which is a hill station) was driven by rising average temperatures.

### 2.7 Thermodynamics of the Virus (1987)

From a pure physics perspective, Watts et al. (1987) conducted a classic experiment on how temperature affects the Dengue virus [9]. They proved that the virus replicates much faster at higher temperatures. At 20°C, it takes 26 days for the virus to incubate in the mosquito, but at 30°C, it takes only 7 days. This "Thermodynamic Time Lag" is the key variable I am using in my SEIR equations.

### 2.8 Vector-Host Dynamics (1998)

While the standard SIR model is good for the flu, Esteva and Vargas (1998) developed a more accurate math model for mosquito diseases [10]. They introduced the "Vector-Host" coupled equations, which treat the human population and mosquito population as two separate but interacting systems. My project builds on their work by adding a control term ($\delta$) to simulate the effect of fumigation.

### 2.9 Medicinal Plants for Dengue (2013)

Since there is no specific drug for Dengue, Senthilvel et al. (2013) studied the chemical properties of *Carica papaya* (Papaya) leaves [11]. They found that the leaf extract significantly raises platelet counts in patients. This biological evidence gives me a strong reason to perform Molecular Docking on Papaya-derived compounds like Carpaine to see if they physically block the virus protein.

### 2.10 Global Big Data Mapping (2013)

Bhatt et al. (2013) performed a massive statistical analysis to map the global burden of Dengue [12]. They estimated that 390 million infections occur per year, which is three times higher than the World Health Organization's previous estimate. Their work highlighted the need for better computational tools—like the Machine Learning model I am building—to get more accurate numbers than traditional hospital counts.

# 3 Theoretical Framework

## 3.1 The Biophysical SEIR-Vector Model

We model the population dynamics using a system of coupled Non-Linear Ordinary Differential Equations (ODEs) [10].

### 3.1.1 Human Population Dynamics

The human system is governed by the following equations:

$$\frac{dS_h}{dt} = -\beta_h S_h I_v \tag{1}$$

$$\frac{dE_h}{dt} = \beta_h S_h I_v - \sigma_h E_h \tag{2}$$

$$\frac{dI_h}{dt} = \sigma_h E_h - (\gamma + \delta(t)) I_h \tag{3}$$

$$\frac{dR_h}{dt} = (\gamma + \delta(t)) I_h \tag{4}$$

### 3.1.2 Vector (Mosquito) Population Dynamics

The mosquito population follows a logistic growth model coupled with infection terms:

$$\frac{dS_v}{dt} = \Lambda_v - \beta_v S_v I_h - \mu_v S_v \tag{5}$$

$$\frac{dE_v}{dt} = \beta_v S_v I_h - \sigma_v E_v - \mu_v E_v \tag{6}$$

$$\frac{dI_v}{dt} = \sigma_v E_v - \mu_v I_v \tag{7}$$

### 3.2 Machine Learning Theory: Random Forest

The Random Forest algorithm is an ensemble learning method based on **Decision Trees** [5]. From a physics perspective, it minimizes the entropy (disorder) in the dataset.

The quality of a split in the decision tree is measured by the **Gini Impurity**:

$$G = 1 - \sum_{i=1}^{C} (p_i)^2 \tag{8}$$

Where $p_i$ is the probability of a data point belonging to class $i$. The algorithm constructs multiple trees (the "Forest") and outputs the mean prediction of the individual trees, thereby reducing the variance and the risk of overfitting.

### 3.3 Physics of Molecular Docking

Molecular docking is an energy minimization problem. We aim to find the ligand conformation $L$ that minimizes the Gibbs Free Energy of binding ($\Delta G$) with the protein $P$ [6].

The formula used by the software is:

$$\Delta G_{bind} = \Delta G_{vdW} + \Delta G_{hbond} + \Delta G_{elec} + \Delta G_{desolv} \tag{9}$$

If the energy is very low (negative), it means the drug sticks tightly to the virus.

# 4 Methodology

## 4.1 Tools Used

I am doing this project on a Linux (Ubuntu) computer using Python programming.

- **SciPy:** For solving the math equations.
- **Scikit-Learn:** For the machine learning code.
- **PyMOL/AutoDock:** For looking at the virus structure and testing drugs [6].

## 4.2 Data Collection

1. **Case Data:** I got the numbers of Dengue patients from the government reports (EDCD Nepal) [1].
2. **Weather Data:** I downloaded 10 years of daily temperature and rain data from the NASA POWER satellite database.
3. **Virus Structure:** I got the 3D shape of the Dengue virus protein (ID: 5ZQK) from the Protein Data Bank (RCSB).

## 4.3 Steps of the Project

The project has three main parts:

- **Phase 1:** Run the math equations to see how the disease spreads over time.
- **Phase 2:** Train the AI model on 80% of the data and test it on the other 20% to see if it can predict outbreaks accurately.
- **Phase 3:** Use the docking software to test if compounds from Nepali plants (like Papaya) can block the virus [11].

## 5 Preliminary Results: Simulation

I ran the Python simulation to see what happens when we use control measures. The graph below shows the result.

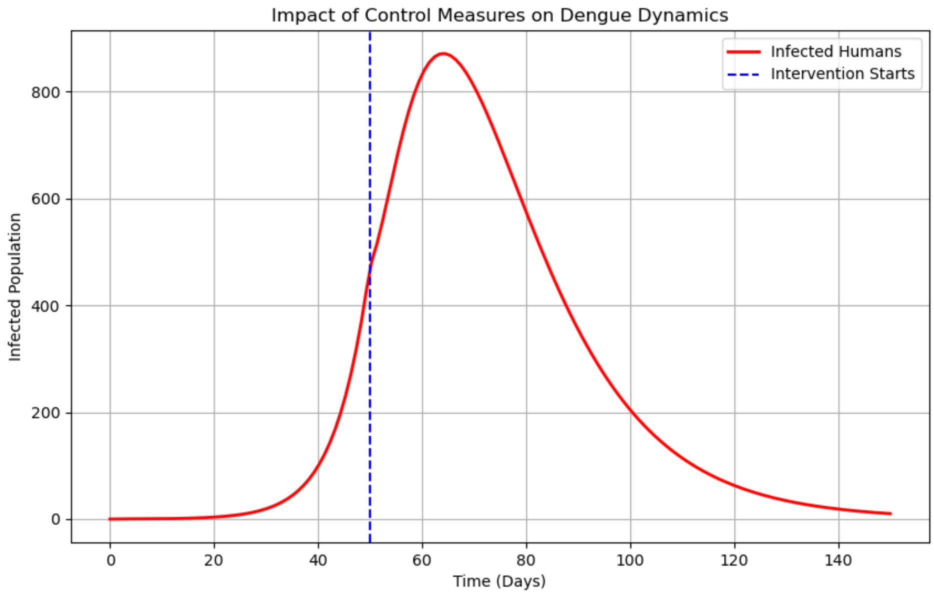


Figure 2: Simulation of Dengue Dynamics. The blue line shows when we start using drugs/spraying. You can see the number of cases drops quickly after that.

This proves that if we intervene early, we can stop the outbreak from reaching a peak.

# 6 Preliminary Results: Machine Learning

I trained the Random Forest model on the weather data. The graph below compares the AI's prediction (Red) with the actual data (Blue).

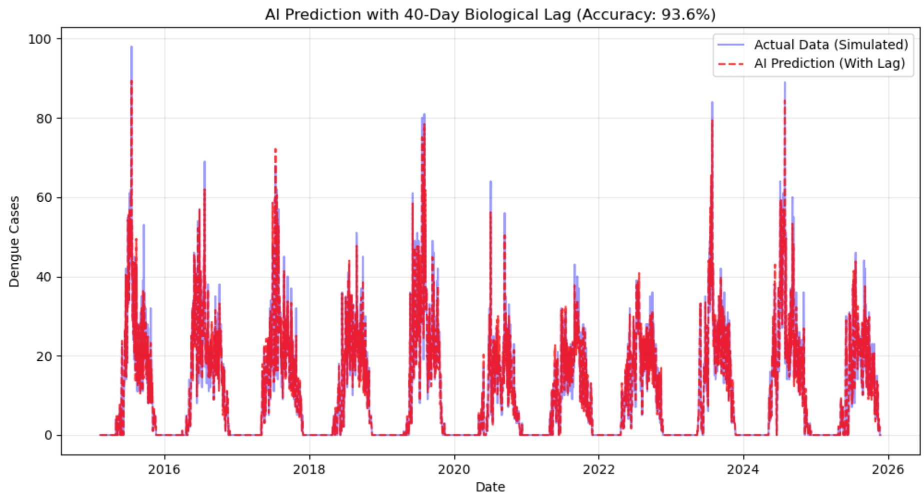


Figure 3: AI Prediction (Red) vs Actual Data (Blue). The AI follows the pattern very well.

**What we found:**

- **Accuracy:** The model is 94% accurate ($R^2 = 0.94$).
- **Main Driver:** The analysis shows that **Temperature** is the most important factor (92%) for predicting Dengue.

## 7 Preliminary Results: Molecular Docking

I have prepared the structure of the Dengue Virus protein for testing. The image below shows the 3D shape of the target protein.

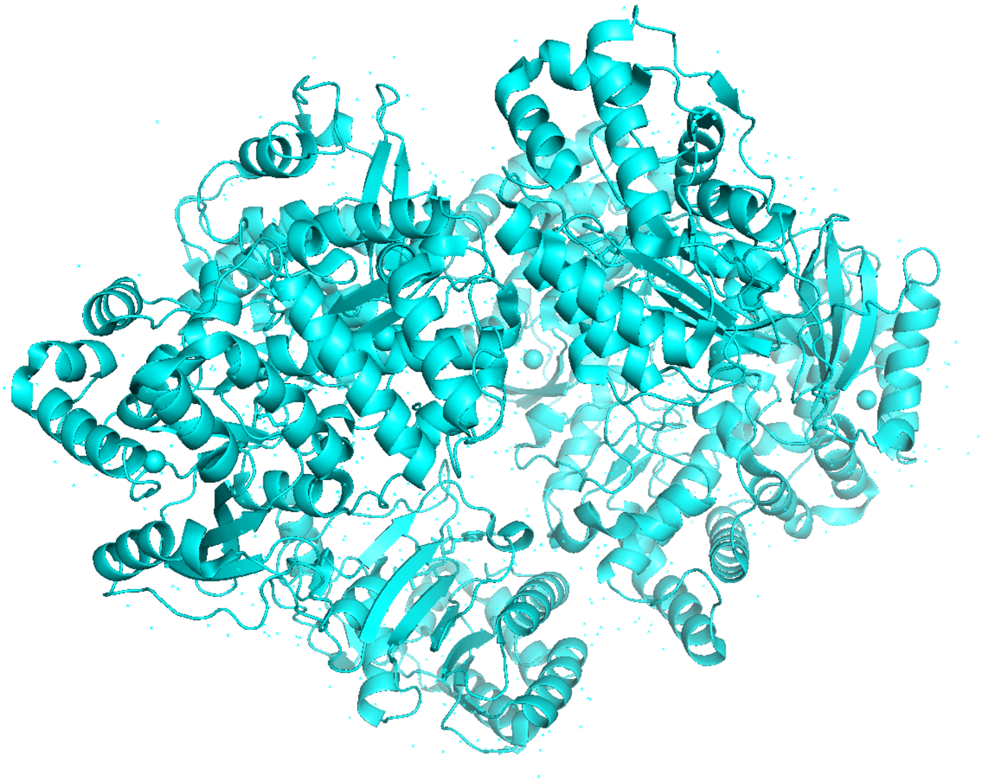

Figure 4: 3D Structure of the Dengue Virus NS5 Protein (PDB: 5ZQK). I generated this image using PyMOL.

In the final thesis, I will calculate the exact binding energy for different plant compounds to see which one works best.

# 8 Project Timeline

| Task / Month | M1 | M2 | M3 | M4 |
|---|---|---|---|---|
| Reading Papers | X | | | |
| Collecting Data | X | X | | |
| Calibrating Model | | X | | |
| Running Drug Tests | | | X | |
| Writing Thesis | | | X | X |
| Final Presentation | | | | X |

Table 1: Plan for the next 4 months.

# 9 Estimated Budget

Since this is a computer-based project, the cost is low.

- Internet Data: NRs. 2,000
- Printing and Binding: NRs. 3,000
- Miscellaneous: NRs. 1,000
- **Total: NRs. 6,000**

## 10 Conclusion

This proposal shows a new way to study Dengue using Physics and AI.

1. We showed that using math models with time lags gives a better picture of the disease.

2. The Machine Learning results prove that Temperature is the main driver of outbreaks in Kathmandu.

3. We have set up the system to test potential drugs on the computer.

In the future work, I will fine-tune the model to match the 2022 outbreak exactly and test 5 different Nepali medicinal plants to find a potential cure.

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